

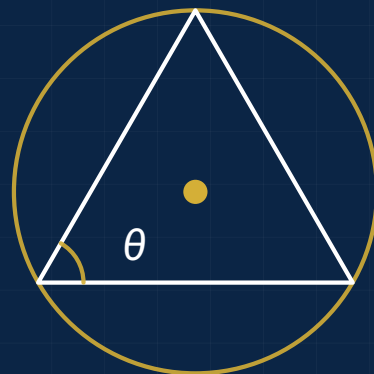


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CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 10

Pythagoras & Trigonometry

In a right-angled triangle: two sides give the third (Pythagoras); a side and an angle give the rest (SOH CAH TOA). Label H, O, A from the angle first.



4

VARIANTS

30

PAST-PAPER PARTS

47

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Pythagoras

V2 SOH CAH TOA

V3 Elevation & depression

V4 Two triangles together

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Right-angled triangles at a glance



Your notes

Pythagoras & Trigonometry

SIDES ONLY

V1 Pythagoras

$$a^2 + b^2 = c^2$$

PAST PAPERS: 6 OF 30

SIDES & ANGLES

V2 SOH CAH TOA

$$\frac{24.3}{\cos 63^\circ}$$

PAST PAPERS: 8 OF 30

PUT IT TOGETHER

V4 Two triangles

shared side

PAST PAPERS: 11 OF 30

THE KEY HABIT

LABEL H, O, A FIRST

From the angle you use; H is opposite the right angle.

V3 Elevation & depression

$$500 \sin 23^\circ$$

PAST PAPERS: 5 OF 30

★ How often does each variant come up?

27 past-paper questions (Jan 2020 → Nov 2025) hold 30 parts; 47 practice parts — grouped by type.

	PAST PAPERS (OUT OF 30)	PRACTICE PARTS
V1 Pythagoras	6	15
V2 SOH CAH TOA	8	16
V3 Elevation, depression	5	4
V4 Two triangles together	11	12

WHAT THIS TELLS YOU On every paper, often twice: a quick one-triangle question early (3 marks) and a two-triangle question later (4–5 marks). The skill is the same: one right-angled triangle at a time.

BEFORE YOU START

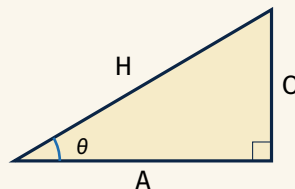
The big idea



Name the sides from the angle.

Hypotenuse: opposite the right angle (longest). **O**pposite: across from your angle. **A**djacent: next to your angle.

Then pick the ratio with the two sides you **have** and **want**.



SOH CAH TOA

SOH

$$\sin \theta = \frac{O}{H}$$

Opposite and hypotenuse.

CAH

$$\cos \theta = \frac{A}{H}$$

Adjacent and hypotenuse.

TOA

$$\tan \theta = \frac{O}{A}$$

Opposite and adjacent.



Three moves

Side on top

$$O = H \sin \theta$$

Multiply.

Side on bottom

$$H = \frac{O}{\sin \theta}$$

Divide.

Angle

$$\theta = \sin^{-1} \frac{O}{H}$$

Inverse key.



CALCULATOR FIRST, THEN CHECK

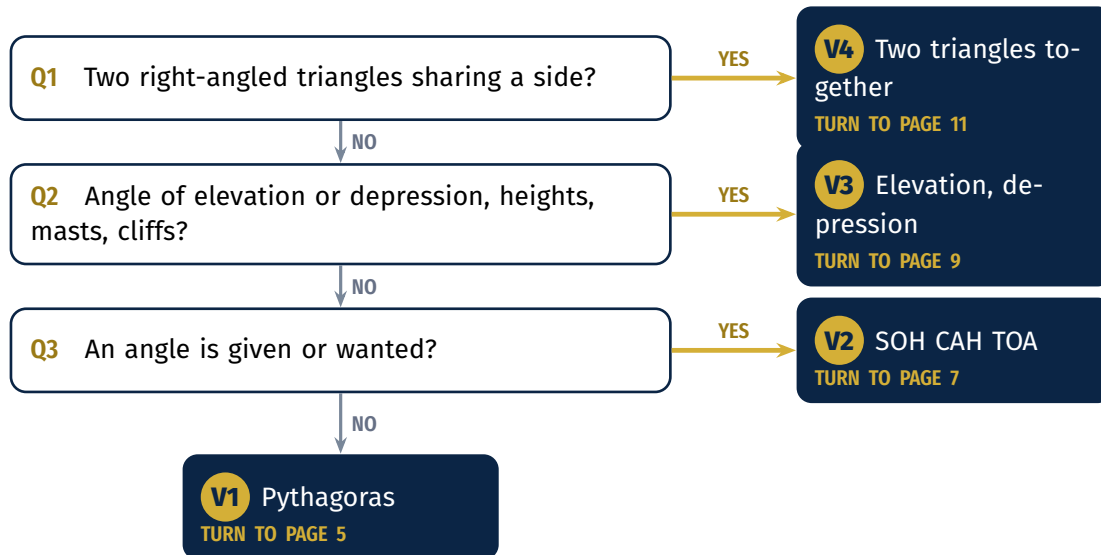
Calculator in degrees (D). Check: the hypotenuse is always the longest side.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?


 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Pythagoras

Longest side: square, add, root. Shorter side: square, subtract, root.

SEEN IN PAST PAPERS

6 / 30

PRACTICE QUESTIONS: 15

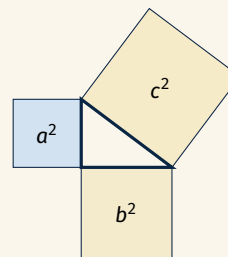

 Your notes

Came up in: Jan 2020 P2HR Q4 • Nov 2021 P1H Q2 • Jan 2022 P1HR Q3 • May 2023 P1H Q5 • May 2024 P1H Q11 • Nov 2025 P2H Q11

? What is it?**In plain words**

In a right-angled triangle, $a^2 + b^2 = c^2$, where **c is the hypotenuse**.

Finding the hypotenuse: **add** the squares. Finding a shorter side: **subtract**. Hidden right triangles: diagonals, heights of isosceles triangles.

**IT LOOKS LIKE**

diagonal of a rectangle

area 240

perimeter

**How do I do it?****1 Right angle**

Find it; the side opposite is c.

2 Which side?

Hypotenuse or a shorter side?

3 Square

Add (for c) or subtract (for a shorter side).

4 Root

Then round; check c is the longest.

LONG: ADD • SHORT: SUBTRACT

Then square root.

V1 Pythagoras

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

“Work out the length of the diagonal of the rectangle” → $\sqrt{a^2 + b^2}$

A side missing and the hypotenuse given → $\sqrt{c^2 - a^2}$

“Area of triangle ABC = 240 ... perimeter” → **Height from the area, then Pythagoras**

 WORKED EXAMPLE **PAST PAPER**

November 2025 · Paper 2H · Q11 · 4 marks

Right-angled triangle ABC, right angle at A; AB = 30 cm; area 240 cm². Work out the perimeter.

STEP 1 $\frac{1}{2} \times 30 \times AC = 240 \Rightarrow AC = 16$ ◀ area

STEP 2 $BC^2 = 30^2 + 16^2 = 1156$ ◀ Pythagoras (hypotenuse)

STEP 3 $BC = 34$ ◀ root

STEP 4 $30 + 16 + 34 = 80$ ◀ perimeter

ANSWER 80 cm

✓ 34 is the longest side ✓

 WATCH OUT

✗ $\sqrt{30^2 - 16^2}$

✓ $\sqrt{30^2 + 16^2}$

BC is the hypotenuse — add.

 EXAM TIP

Write $a^2 + b^2 = c^2$ with the numbers in before you press anything.

 Check yourself

calculator first · answers on page 13

1.1 Jan 2020 P2HR

A rectangle is 8.5 cm by 5.6 cm. Length of its diagonal (1 d.p.)?

Answer:

1.2 Nov 2021 P1H

Right-angled triangle: hypotenuse 52 cm, one side 30 cm, other side h . Find h (3 s.f.).

Answer:

1.3 Dr Eslam

The shorter sides are 7 cm and 24 cm. Hypotenuse?

Answer:

V2

VARIANT 2

SOH CAH TOA

Label O, A, H; choose the ratio; multiply, divide, or use the inverse key.

SEEN IN PAST PAPERS

8 / 30

PRACTICE QUESTIONS: 16


 Your notes

Came up in: Jan 2020 P1H Q9 • Jan 2021 P1H Q10 • May 2021 P2H Q7 • Jan 2022 P1H Q3 • Jun 2022 P2H Q6 • Jun 2024 P2H Q5 • Nov 2024 P2H Q3 • Jun 2025 P1HR Q9

? What is it?

In plain words

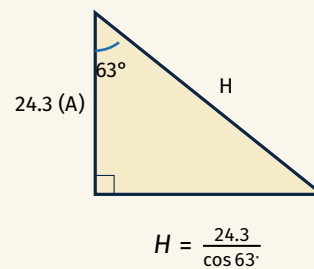
Stand at the angle. Label the sides **O, A, H**. Pick the ratio that uses the side you know and the side you want. Unknown on **top**: multiply. Unknown on the **bottom**: divide. Angle: **inverse**.

IT LOOKS LIKE

find PQ

find x

find the angle



How do I do it?

1 Label

H opposite the right angle; O, A from the angle.

2 Choose

SOH, CAH or TOA with your two sides.

3 Rearrange

Top: $\times$. Bottom: $\div$. Angle: $^{-1}$.

4 Round

To the accuracy asked.

LABEL, CHOOSE, SOLVE Degrees mode.

V2 SOH CAH TOA

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

"Calculate the length of PQ" with an angle → **Label; pick the ratio**

"Work out the size of angle ..." → $\sin^{-1}, \cos^{-1}, \tan^{-1}$

Isosceles triangle with a perpendicular → **Two right triangles: half the base**

WORKED EXAMPLE PAST PAPER

January 2020 · Paper 1H · Q9 · 3 marks

Right-angled triangle PQR with the right angle at R ; $PR = 24.3$ cm and angle $QPR = 63^\circ$. Calculate PQ (3 s.f.).

STEP 1 $PR = A, PQ = H$ ◀ from the 63° angle

STEP 2 $\cos 63^\circ = \frac{24.3}{PQ}$ ◀ CAH

STEP 3 $PQ = \frac{24.3}{\cos 63^\circ}$ ◀ unknown on the bottom: divide

STEP 4 $= 53.52 \dots$ ◀

ANSWER 53.5 cm

✓ longer than 24.3 (hypotenuse) ✓

WATCH OUT

× $PQ = 24.3 \cos 63^\circ$

✓ $PQ = \frac{24.3}{\cos 63^\circ}$

PQ is on the bottom of the fraction — divide.

EXAM TIP

Write the ratio with the letters first ($\cos = \frac{A}{H}$), then the numbers.

Check yourself

calculator first · answers on page 13

2.1 Jan 2022 P1H

Right angle at B ;
 $AC = 8.4$ cm; angle
 $ACB = 65^\circ$. Find AB (3
s.f.).

Answer:

2.2 Jun 2025 P1HR

Right angle at Q ;
 $PQ = 6.5$ cm; angle
 $R = 24^\circ$. Find QR (3 s.f.).

Answer:

2.3 Dr Eslam

A right-angled triangle
has opposite 5 cm and
adjacent 12 cm. Find
the angle.

Answer:

V3

VARIANT 3

Elevation & Depression

Both are measured from the horizontal;
draw the right triangle, then SOH CAH TOA.

SEEN IN PAST PAPERS

5 / 30

PRACTICE QUESTIONS: 4


Your notes

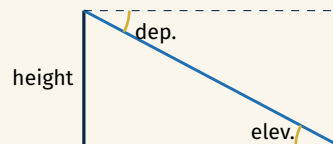
Came up in: Jan 2021 P2HR Q13 • Nov 2021 P1H Q12 • Jan 2022 P2HR Q12 • Jan 2022 P2H Q14

? What is it?

In plain words

Elevation: look **up** from the horizontal. **Depression:** look **down** from the horizontal.

They are equal (alternate angles), so the angle goes **inside** the right triangle at the bottom.



IT LOOKS LIKE

balloons masts
cliff and buoy

How do I do it?

1 Horizontal

Draw it; the angle starts there.

2 Triangle

Height, horizontal distance, line of sight.

3 Ratio

tan for height and distance; sin/cos with the sight line.

4 Extra

Add or subtract other heights (posts, cliffs).

FROM THE HORIZONTAL Never from the vertical.

V3 Elevation & Depression

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"The angle of depression of B from A is 23° " → **Same angle at B (alternate)**

"two vertical phone masts ... angle of depression of B from D" → **Height difference ÷ distance**

"least possible height of the post" → **Smallest allowed angle**

WORKED EXAMPLE **PAST PAPER**

January 2021 · Paper 2HR · Q13 · 3 marks

Points A and B on two balloons are 500 m apart in a straight line. The angle of depression of B from A is 23° . Calculate the vertical height of A above B (1 d.p.).

STEP 1 $H = 500$, height = 0 ◀ angle 23° at B (alternate)

STEP 2 $\sin 23^\circ = \frac{h}{500}$ ◀ SOH

STEP 3 $h = 500 \sin 23^\circ$ ◀ multiply

STEP 4 $= 195.36 \dots$ ◀

ANSWER 195.4 m

✓ less than 500 ✓

WATCH OUT✗ using $\cos 23^\circ$ for the height✓ $\sin 23^\circ$ The height is opposite the 23° angle.**EXAM TIP**Sketch the horizontal line through A and mark the 23° there, then copy it to B.**Check yourself**

calculator first · answers on page 13

3.1 Jan 2022 P2HR

Masts AB = 6.2 m and CD = 30.7 m stand 244 m apart. Angle of depression of B from D?

Answer:

3.2 Nov 2021 P1H

A cliff is 100 m high; from a buoy d m away the elevation of the top is 20° . Find d (3 s.f.).

Answer:

3.3 Jan 2022 P2H

Zip wire from post AB to post CD = 2.6 m, 12 m apart, at least 5° to the horizontal. Least height of AB?

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

Two Triangles Together

Solve the triangle with enough information first; the shared side carries you into the second.

SEEN IN PAST PAPERS

11 / 30

PRACTICE QUESTIONS: 12


 Your notes

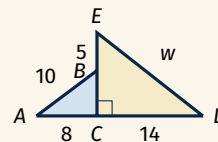
Came up in: Jan 2020 P1HR Q15 • Jun 2022 P2HR Q10 • Jun 2023 P2H Q12 • Nov 2023 P1H Q10 • Nov 2023 P2H Q2 • Nov 2024 P1H Q12 • Jun 2025 P2HR Q11 • Jun 2025 P2H Q10 • Nov 2025 P1H Q11

? What is it?

In plain words

Two right-angled triangles share a side (or sit along one line). One of them has **two facts** — solve it first.

The **shared side** becomes a known side of the second triangle. Then subtract or add lengths along the line.



IT LOOKS LIKE

shared height

points on one line

How do I do it?

1 Start

The triangle with two known facts.

2 Shared side

Find it; mark it on the diagram.

3 Second triangle

Pythagoras or trig with the shared side.

4 Combine

Add or subtract lengths/angles as asked.

SHARED SIDE CARRIES YOU The most common variant here.

V4 Two Triangles Together

VARIANT 4 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

“ACD and EBC are straight lines” → **Lengths add along the lines**

“D is a point on BC ... find DC” → **BC – BD**

“Work out the size of angle ACD” → **Angle difference of two triangles**

WORKED EXAMPLE PAST PAPER

November 2023 · Paper 2H · Q2 · 4 marks

Right angles at C. AB = 10 cm, AC = 8 cm; EB = 5 cm; CD = 14 cm; ACD and EBC are straight. Find ED = w (1 d.p.).

STEP 1 $BC = \sqrt{10^2 - 8^2} = 6$ ◀ first triangle

STEP 2 $EC = 6 + 5 = 11$ ◀ along the line

STEP 3 $w^2 = 11^2 + 14^2 = 317$ ◀ second triangle

STEP 4 $w = 17.80 \dots$ ◀ root

ANSWER $w = 17.8$

✓ longest side of triangle ECD ✓

WATCH OUT

✗ $w^2 = 5^2 + 14^2$

✓ $w^2 = 11^2 + 14^2$

EC is the whole vertical side, 5 + 6.

EXAM TIP

Write every length you find on the diagram — the second triangle appears.

Check yourself

calculator first · answers on page 13

4.1

Nov 2024 P1H

Right angle at B;
AB = 4250 m; angle
BAD = 47°; angle
BCA = 24°; D on BC.
Find DC.

Answer:

4.2

Jun 2025 P2H

Right angles at B:
AB = 16, BC = 12,
AD = 1.5 × AC, C on BD.
Find CD (3 s.f.).

Answer:

4.3

Nov 2023 P1H

Height from C is 8 m;
angle A = 40°; AB = 22
m. Find angle B = x° (1
d.p.).

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked in Python (check_answers.py).

1.1 10.2 cm**1.2** $h = 42.5$ **1.3** 25 cm**2.1** 7.61 cm**2.2** 14.6 cm**2.3** 22.6° **3.1** 5.7° **3.2** $d = 275$ **3.3** 3.65 m**4.1** 4988 m**4.2** 13.4 cm**4.3** $x = 32.7$

THE WHOLE TOPIC ON ONE PAGE

Triangles cheat sheet



Your notes
V1 Pythagoras 6/30

$$a^2 + b^2 = c^2$$

Long: add; short: subtract.

V2 SOH CAH TOA 8/30

$$\frac{24.3}{\cos 63^\circ}$$

Label, choose, solve.

V3 Elevation 5/30

$$500 \sin 23^\circ$$

From the horizontal.

V4 Two triangles 11/30

$$6 + 5 = 11$$

Shared side carries you.

**TOP 3 LOST MARKS**

1. multiplying when the unknown is on the bottom ×
2. adding squares for a short side ×
3. radians mode ×

BOTTOM LINE Find the right angle · label H, O, A · one triangle at a time · degrees mode.